

## Overview

Our implementation simulates particle movement and collision detection within a bounded square area. The algorithm uses a grid-based spatial partitioning strategy to efficiently manage particle interactions by limiting collision checks to nearby particles within the same or adjacent grid cells. A `GridCell` is implemented as a vector of pointers to particles within the grid cell and the `Grid` is implemented as a 2D vector of `GridCells` representing the simulation space. The grid cell size is set as a multiple of the particle radius.

For each timestep, our algorithm goes through 3 main phases (See [Appendix A](#)):

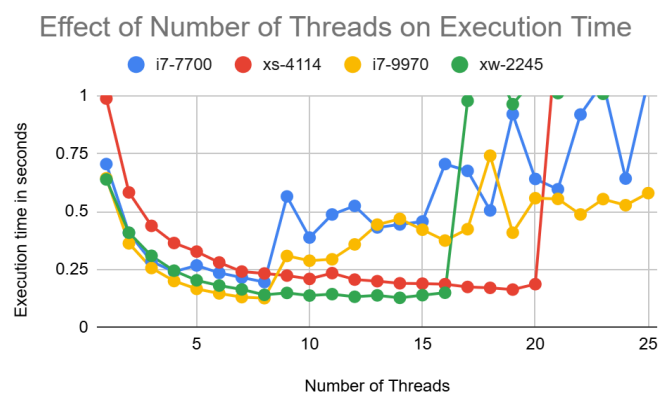
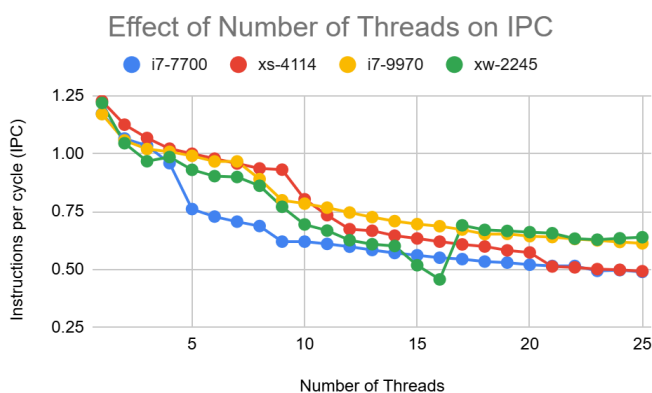
1. Position Update: Update particle positions based on their velocities
2. Reassignment of particles to `GridCells`
3. Velocity Update: Collision Detection and Resolution with walls and other particles

The grid structure allows for efficient parallel processing by dividing the work among threads. OpenMP is used to parallelize all 3 phases:

- `#pragma omp parallel for` is used to parallelize loops for updating particle positions and clearing grid cells, where each thread independently updates each particle's position and clears each grid cell.
- `#pragma omp parallel for collapse(2) reduction(| : collided)` is used to parallelize nested loops for the velocity update phase, where each thread independently handles collision detection and resolution for each grid cell. The reduction ensures that the `collided` variable is correctly updated across all threads

To avoid data races from different threads handling the same particle when resolving collisions between particles from adjacent grid cells, the grid is divided into 2x2 subregions. The quadrants for each subregion are processed sequentially while each of the 4 quadrants for all subregions are processed in parallel (i.e. all top left first, then all bottom left...).

Due to parallelization, as the number of threads increases, the workload is distributed more, decreasing the number of instructions per cycle (IPC), hence reducing the time required for each timestep, lowering its execution time. Beyond the number of hardware cores (8 for i7-7700 and i7-9970, 16 for xw-2245, 20 for i7-9970), there is an increase in execution time instead due to overheads from parallelization.

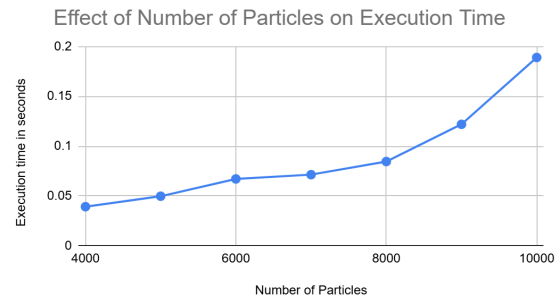
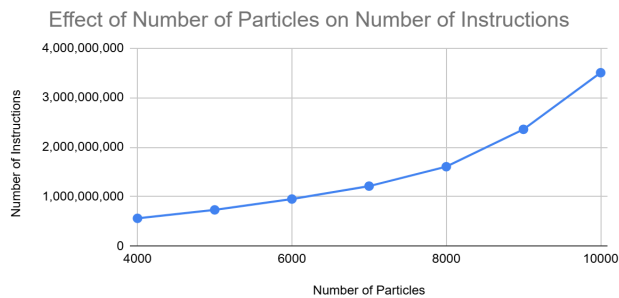


## Performance Analysis

The program's performance would be influenced by parameters specified in the input file:

### Number of Particles (**param\_particles**)

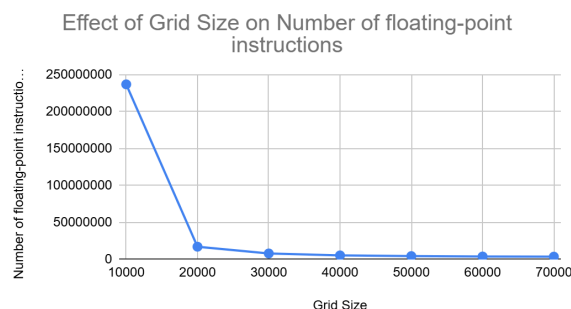
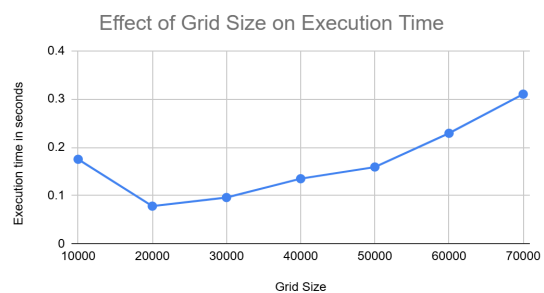
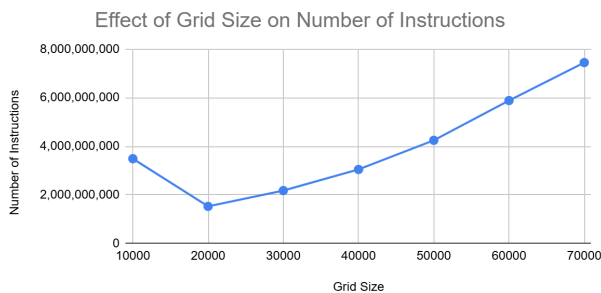
As the number of particles increases, the particle density increases, increasing the number of calculations for updating positions and handling collisions are required, increasing the computational load, hence increasing the number of instructions and execution time. The increase is more than linear as the number of possible collisions between particles to be checked and resolved increases more than linearly.



### Grid size (**square\_size**)

Initially, as the grid size increases, the number of particles per cell decreases, improving collision detection efficiency, reducing the number of double-precision floating-point instructions consistently. However, as the grid cell size is defined as a multiple of particle radius, a larger grid size also increases the number of grid cells to manage. As such, beyond that, the overheads from managing the grid cells outweigh the increase in performance from fewer collisions.

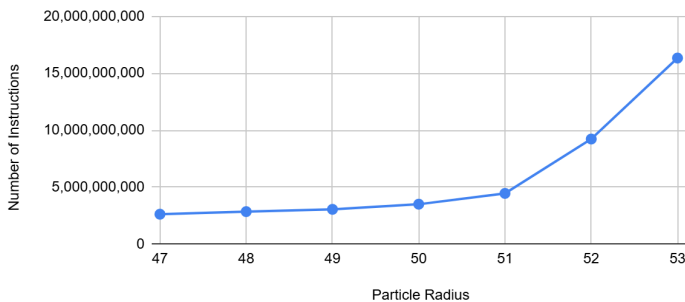
Given that the performance requirements have a minimum particle density of 0.7, we can keep the grid cell size to be defined as a multiple of particle radius.



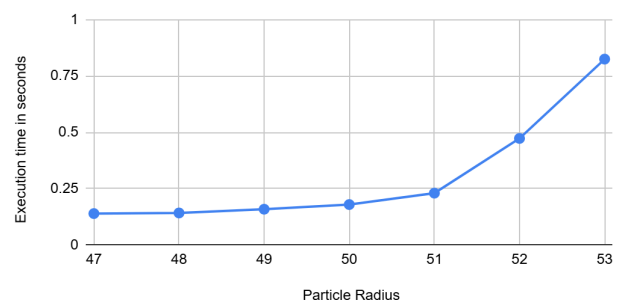
### Particle Radius (**param\_radius**)

As particle radius increases, the cell size increases, reducing the number of grid cells to process but also increasing the number of particles in each cell and hence the number of collision checks within each cell. Initially, both these two factors roughly balance out as the number of instructions and execution time only increases slightly as particle radius increases from 47 to 50. However, as particle radius increases beyond 50, the particle density increases significantly more than the reduction in number of grid cells to process, causing the number of instructions and execution time to spike upwards.

Effect of Particle Radius on Number of Instructions



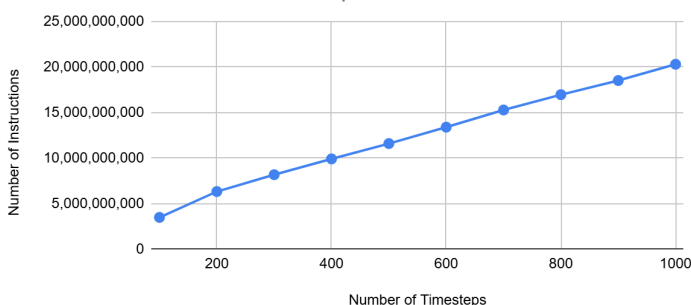
Effect of Particle Radius on Execution Time



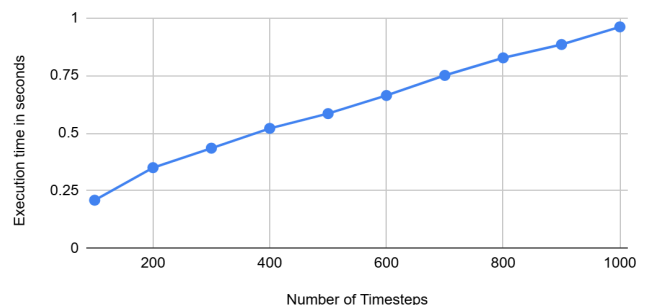
### Number of Steps (**param\_steps**)

Measurements were taken with the particles having the same starting locations and velocities. As the number of steps increases, the workload increases roughly linearly, increasing the number of instructions and execution time linearly.

Effect of Number of Timesteps on Number of Instructions

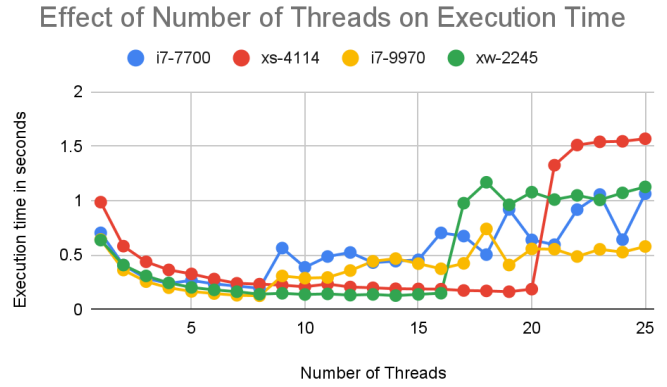
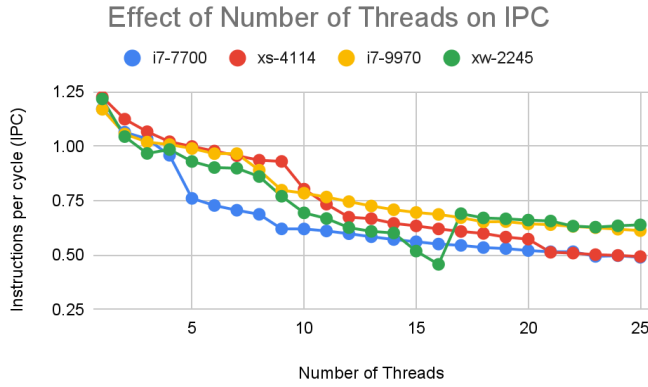


Effect of Number of Timesteps on Execution Time



### Hardware Type

The performance of the program would also be influenced by the hardware type of the machine the program is run on. As mentioned above, due to parallelization, as the number of threads increases, the workload is distributed more, decreasing the number of instructions per cycle (IPC), reducing the time required for each timestep, lowering its execution time. Beyond the number of hardware cores (8 for i7-7700 and i7-9970, 16 for xw-2245, 20 for i7-9970), execution time increases instead due to overheads from parallelization.

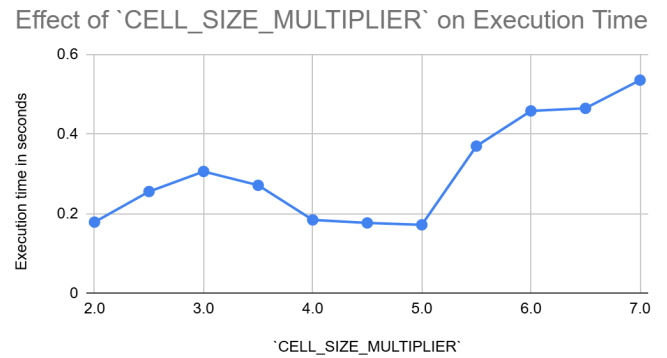
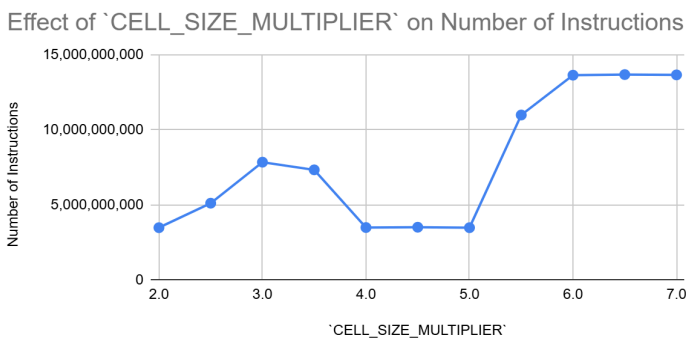


## Optimizations

### Varying Cell Size Multiplier

The grid cell size is defined as a multiple of particle radius (`CELL_SIZE_MULTIPLIER`). Varying the `CELL_SIZE_MULTIPLIER` can optimize the balance between the number of grid cells and the number of particles per cell. A larger multiplier reduces the number of grid cells, potentially decreasing the overheads of managing the grid, but increases the number of particles per cell, which may increase the number of collision checks. Conversely, a smaller multiplier increases the number of grid cells, reducing the number of particles per cell but increasing the grid management overhead.

For correctness, the `CELL_SIZE_MULTIPLIER` value has to be minimally 2 since only the adjacent grid cells are checked for detecting collisions. The `CELL_SIZE_MULTIPLIER` was varied from 2.0 to 7.0 in increments of 0.5, and the number of instructions execution time was measured for each value.



From our measurements, we note that a `CELL_SIZE_MULTIPLIER` of 2.0 or between 4.0 or 5.0 would be optimal to maximise the performance of our program.

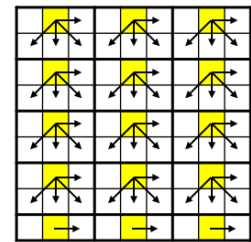
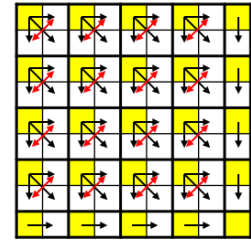
### Minimizing Contention when Parallelizing Grid Traversal

When updating velocity for particle-particle collisions, a naive approach that processes each grid cell independently would lead to data races when checking adjacent cells. One potential solution is to use locks to prevent conflicts. However, this approach was discarded due to

high contention - adjacent cells would frequently attempt to lock each other - along with the complexity of managing deadlocks when acquiring multiple locks.

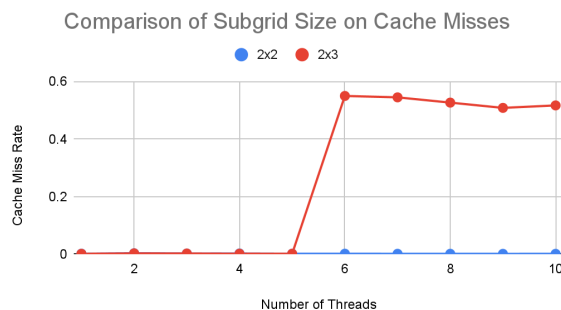
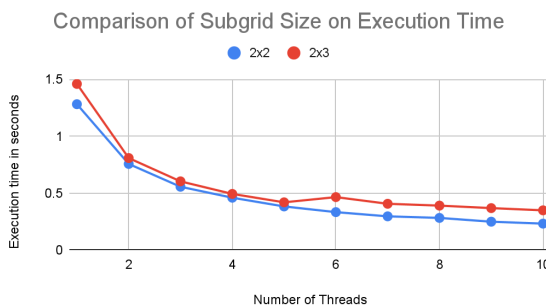
Instead, a lock-free approach was adopted by processing data race prone regions sequentially. 2 methods were considered:

1. Processing  $2 \times 2$  cell chunks: The primary comparison occurs between the target cell (yellow) and its adjacent cells, with an additional check required between two adjacent cells (red arrow). This results in 4 sequential runs.
2. Processing  $2 \times 3$  cell chunks: This method addresses the edge case from the  $2 \times 2$  approach by adding an extra check for the target cell, ensuring all necessary comparisons are covered. This results in 6 sequential runs.



When comparing the results of both runs, the  $2 \times 2$  subgrid cell chunk size performs faster at all thread numbers, with a higher deviation beyond 5 cores. This can be attributed to a significantly higher cache miss rate at 6 cores and above, reaching close to 50%.

This performance difference is likely because accessing a  $2 \times 3$  subgrid in parallel leads to more cache misses and a slower program compared to a  $2 \times 2$  subgrid. A  $2 \times 3$  subgrid accesses more data per thread, leading to a much higher cache miss rate once the cache is full. A similar behavior is expected for a CPU with more cores and threads on both algorithms. However the  $2 \times 3$  subgrid is more prone to cache thrashing as compared to the  $2 \times 2$  subgrid.



## Updating of Grid Data

In the initial implementation, the grid was updated at each step sequentially to reassign each particle to a grid cell. An optimization we tried was to create a **vector** of updates to each cell whenever a particle changes cell. However due to the dense spread of the particles, the cost to update each grid cell was significantly higher than statically reassigning the entire grid. Even an implementation that involved **unordered\_set** provided no improvement - in fact the overhead of the **unordered\_set** caused the execution time to increase significantly as compared to the statically accessed **vector**.

## Appendix

### Appendix A: Code Implementation

High-level overview of steps taken in each timestep:

1. Position Update
2. Reassignment of particles to grid cells
3. Velocity Update: Collision Detection and Resolution

```
// For each timestep
while (params.param_steps--) {
    // Phase I: Position Update - Move each particle based on velocity
    position_update(particles);

    // Reassign particles to new grid cells
    update_grid(grid, particles, cell_size, grid_width, grid_height);

    // Phase II: Velocity Update: Collision Detection and Resolution
    bool collided;
    const int interval_i = 2;
    const int interval_j = 2;
    do {
        collided = velocity_update_wall(grid, grid_width, grid_height, params);

        for (int i = 0; i < interval_i; ++i) {
            for (int j = 0; j < interval_j; ++j) {
                collided |=
                    velocity_update_particle(grid, grid_width, grid_height, params, i,
j, interval_i, interval_j);
            }
        }
    } while (collided);
}
```

## Appendix B: Performance Analysis

The raw data `slurmlog` files can be found in the folder [here](#)

The spreadsheet with the processed data can be found [here](#).

### Variables:

Measurements for the following variables were taken. The ranges for the variables measured are given below (If the variables are not being tested, they are kept at the constant value specified unless otherwise mentioned):

- Number of Threads: 1-25 (Constant 8)
- Hardware Type: xs-4114, xw-2245, i7-7700, i7-9700 (Constant: i7-7700)
- Cell Size Multiplier: 2.0-7.0 (Constant 2.0)
- Number of Particles: 4K-10K (Constant: 10K)
- Grid/Square Size: 10K-70K (Constant: 10K)
- Particle Radius: 47-53 (Constant: 50)
- Number of Timesteps: 100-1000 (Constant: 100)

## Sample Script

The following is an example of the script used to run our program and collect measurements for performance analysis. The partition, `INPUT_FILE_PATH` and `NUM_OF_THREADS` are varied as necessary based on the parameters being tested. 3 measurements were taken to get an average value.

The below sample script was used for testing the performance when varying the number of threads. The other scripts used for testing can be found in our GitHub Classroom repository.

```
#!/bin/bash

## This is an example Slurm template job script for A1 that just runs the script and
arguments you pass in via `srun`.

#SBATCH --job-name=test_num-threads
#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --mem=4gb
#SBATCH --partition=i7-7700
#SBATCH --time=00:10:00
#SBATCH --output=%x_%p_%j.slurmlog
#SBATCH --error=%x_%p_%j.slurmlog

INPUT_FILE_PATH=./tests/standard/10k_density_0.8.in

echo "We are running on $(hostname)"
echo "Job started at $(date)"
echo "Partition: $SLURM_JOB_PARTITION"
echo "Input file path: $INPUT_FILE_PATH"

# Compile your code in case you forgot
echo "Compiling..."
srun make

# Runs your script with the arguments you passed in
echo "Running..."

for NUM_OF_THREADS in {1..10} ## Change this to the number of threads you want to test
do
    echo "Running with $NUM_OF_THREADS threads..."
    srun perf stat -r 3 -e
instructions,cycles,fp_arith_inst_retired.scalar_double,fp_arith_inst_retired.128b_packed_
double ./sim.perf $INPUT_FILE_PATH $NUM_OF_THREADS
done

echo "Job ended at $(date)"
```



### Results from varying Number of Threads on Intel Core i7-7700 machine

For ease of organisation and to avoid timeout, batch script was submitted in 3 batches (1-10 threads, 11-20 threads, 21-25 threads)

The batch for 1-10 threads was run on pdc-015.

The batch for 11-20 threads was run on pdc-012.

The batch for 21-25 threads was run on pdc-013.

| Number of Threads | Number of Instructions | Number of Cycles | Number of Cycles (Error) | Instructions per cycle (IPC) | Execution Time in seconds | Execution Time (Error) | Number of retired scalar double-precision floating-point instructions | Number of retired 128-bit packed double-precision floating-point instructions | Total number of double-precision floating-point instructions | Number (in millions) of floating-point operations per second (MFLOPS) |
|-------------------|------------------------|------------------|--------------------------|------------------------------|---------------------------|------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------|
| 1                 | 3,436,683,878          | 2,932,325,875    | 0.30%                    | 1.17                         | 0.704970                  | 0.35%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 335.5771934                                                           |
| 2                 | 3,452,508,720          | 3,242,369,616    | 0.72%                    | 1.06                         | 0.408380                  | 0.96%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 579.2934375                                                           |
| 3                 | 3,458,354,692          | 3,343,638,285    | 0.09%                    | 1.03                         | 0.284824                  | 0.33%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 830.5896062                                                           |
| 4                 | 3,469,648,196          | 3,618,585,680    | 1.53%                    | 0.96                         | 0.241070                  | 4.01%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 981.3409134                                                           |
| 5                 | 3,500,278,907          | 4,600,395,220    | 0.39%                    | 0.76                         | 0.267800                  | 9.43%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 883.3900448                                                           |
| 6                 | 3,495,869,780          | 4,800,286,574    | 0.26%                    | 0.73                         | 0.235900                  | 10.85%                 | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 1,002.8480458                                                         |
| 7                 | 3,492,290,355          | 4,943,976,467    | 0.50%                    | 0.71                         | 0.216000                  | 11.45%                 | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 1,095.2400648                                                         |
| 8                 | 3,490,855,981          | 5,076,873,042    | 0.15%                    | 0.69                         | 0.196300                  | 13.21%                 | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 1,205.1546307                                                         |
| 9                 | 4,007,262,069          | 6,454,563,706    | 0.65%                    | 0.62                         | 0.565000                  | 32.51%                 | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 418.7112460                                                           |
| 10                | 4,075,819,957          | 6,569,886,911    | 0.04%                    | 0.62                         | 0.388020                  | 2.21%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 609.6898459                                                           |
| 11                | 4,162,773,120          | 6,811,543,571    | 0.33%                    | 0.61                         | 0.487500                  | 18.23%                 | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 485.2755979                                                           |
| 12                | 4,263,766,745          | 7,123,961,912    | 0.08%                    | 0.60                         | 0.524550                  | 0.75%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 450.9996263                                                           |

|    |               |                |       |      |          |        |             |            |             |             |
|----|---------------|----------------|-------|------|----------|--------|-------------|------------|-------------|-------------|
| 13 | 4,381,490,479 | 7,493,797,508  | 0.03% | 0.58 | 0.431400 | 2.81%  | 189,055,270 | 47,516,584 | 236,571,854 | 548.3816736 |
| 14 | 4,502,448,020 | 7,874,711,382  | 0.03% | 0.57 | 0.444640 | 1.36%  | 189,055,270 | 47,516,584 | 236,571,854 | 532.0525684 |
| 15 | 4,613,834,697 | 8,216,495,737  | 0.08% | 0.56 | 0.457130 | 0.55%  | 189,055,270 | 47,516,584 | 236,571,854 | 517.5154857 |
| 16 | 4,728,376,120 | 8,581,454,046  | 0.14% | 0.55 | 0.704700 | 2.64%  | 189,055,270 | 47,516,584 | 236,571,854 | 335.7057670 |
| 17 | 4,835,299,446 | 8,873,785,025  | 0.31% | 0.54 | 0.675300 | 12.37% | 189,055,270 | 47,516,584 | 236,571,854 | 350.3211225 |
| 18 | 4,917,760,585 | 9,194,380,214  | 0.12% | 0.53 | 0.504900 | 2.15%  | 189,055,270 | 47,516,584 | 236,571,854 | 468.5518994 |
| 19 | 5,054,159,638 | 9,530,540,274  | 0.65% | 0.53 | 0.920000 | 23.33% | 189,055,270 | 47,516,584 | 236,571,854 | 257.1433196 |
| 20 | 5,155,256,175 | 9,892,020,245  | 0.06% | 0.52 | 0.640500 | 10.76% | 189,055,270 | 47,516,584 | 236,571,854 | 369.3549633 |
| 21 | 5,270,622,200 | 10,222,794,631 | 0.08% | 0.52 | 0.596260 | 1.10%  | 189,055,270 | 47,516,584 | 236,571,854 | 396.7595579 |
| 22 | 5,402,305,042 | 10,473,930,797 | 0.49% | 0.52 | 0.918500 | 8.16%  | 189,055,270 | 47,516,584 | 236,571,854 | 257.5632597 |
| 23 | 5,492,010,858 | 11,110,016,991 | 0.74% | 0.49 | 1.057000 | 17.08% | 189,055,270 | 47,516,584 | 236,571,854 | 223.8144314 |
| 24 | 5,582,147,261 | 11,218,737,338 | 0.10% | 0.50 | 0.642370 | 1.36%  | 189,055,270 | 47,516,584 | 236,571,854 | 368.2797360 |
| 25 | 5,696,815,593 | 11,623,277,549 | 0.04% | 0.49 | 1.064620 | 0.49%  | 189,055,270 | 47,516,584 | 236,571,854 | 222.2124833 |

### Results from varying Number of Threads on Intel Core xs-4114 machine

For ease of organisation and to avoid timeout, batch script was submitted in 3 batches (1-10 threads, 11-20 threads, 21-25 threads)

The batch for 1-10 threads was run on pdc-008.

The batch for 11-20 threads was run on pdc-004.

The batch for 21-25 threads was run on pdc-004.

| Number of Threads | Number of Instructions | Number of Cycles | Number of Cycles (Error) | Instructions per cycle (IPC) | Execution Time in seconds | Execution Time (Error) | Number of retired scalar double-precision floating-point instructions | Number of retired 128-bit packed double-precision floating-point instructions | Total number of double-precision floating-point instructions | Number (in millions) of floating-point operations per second (MFLOPS) |
|-------------------|------------------------|------------------|--------------------------|------------------------------|---------------------------|------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------|
| 1                 | 3,436,938,932          | 2,802,493,303    | 0.39%                    | 1.23                         | 0.986940                  | 0.75%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 239.7023669                                                           |
| 2                 | 3,450,633,896          | 3,067,853,382    | 0.45%                    | 1.12                         | 0.582680                  | 1.63%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 406.0064770                                                           |
| 3                 | 3,460,030,107          | 3,242,060,371    | 0.48%                    | 1.07                         | 0.438900                  | 2.86%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 539.0108316                                                           |
| 4                 | 3,468,882,790          | 3,396,564,225    | 0.10%                    | 1.02                         | 0.364620                  | 1.00%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 648.8175470                                                           |
| 5                 | 3,475,450,244          | 3,477,840,739    | 0.45%                    | 1.00                         | 0.327600                  | 4.62%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 722.1363065                                                           |
| 6                 | 3,481,338,969          | 3,558,242,947    | 0.87%                    | 0.98                         | 0.280570                  | 2.79%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 843.1829989                                                           |
| 7                 | 3,488,388,016          | 3,644,527,670    | 1.16%                    | 0.96                         | 0.241610                  | 3.75%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 979.1476098                                                           |
| 8                 | 3,495,488,212          | 3,733,520,995    | 0.89%                    | 0.94                         | 0.233590                  | 2.50%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 1,012.7653324                                                         |
| 9                 | 3,500,852,660          | 3,761,939,876    | 5.86%                    | 0.93                         | 0.224300                  | 5.13%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 1,054.7117878                                                         |
| 10                | 3,527,922,488          | 4,391,959,027    | 5.33%                    | 0.80                         | 0.209850                  | 3.72%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 1,127.3378794                                                         |
| 11                | 3,546,542,679          | 4,830,475,895    | 1.75%                    | 0.73                         | 0.234600                  | 14.03%                 | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 1,008.4051748                                                         |
| 12                | 3,557,785,251          | 5,275,929,203    | 0.88%                    | 0.67                         | 0.207240                  | 3.19%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 1,141.5356784                                                         |
| 13                | 3,553,785,534          | 5,322,064,350    | 0.38%                    | 0.67                         | 0.200090                  | 4.61%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 1,182.3272227                                                         |
| 14                | 3,555,334,744          | 5,502,896,670    | 0.44%                    | 0.65                         | 0.191260                  | 4.82%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 1,236.9123392                                                         |

|    |               |                |        |      |          |        |             |            |             |               |
|----|---------------|----------------|--------|------|----------|--------|-------------|------------|-------------|---------------|
| 15 | 3,555,703,123 | 5,604,196,461  | 0.18%  | 0.63 | 0.189900 | 8.64%  | 189,055,270 | 47,516,584 | 236,571,854 | 1,245.7706898 |
| 16 | 3,556,306,982 | 5,730,920,258  | 0.33%  | 0.62 | 0.187450 | 3.55%  | 189,055,270 | 47,516,584 | 236,571,854 | 1,262.0531022 |
| 17 | 3,558,107,259 | 5,843,576,199  | 1.31%  | 0.61 | 0.175310 | 3.74%  | 189,055,270 | 47,516,584 | 236,571,854 | 1,349.4487137 |
| 18 | 3,560,442,508 | 5,935,848,209  | 0.73%  | 0.60 | 0.171600 | 6.05%  | 189,055,270 | 47,516,584 | 236,571,854 | 1,378.6238578 |
| 19 | 3,564,444,176 | 6,113,912,482  | 0.11%  | 0.58 | 0.164100 | 6.22%  | 189,055,270 | 47,516,584 | 236,571,854 | 1,441.6322608 |
| 20 | 3,566,182,706 | 6,218,641,451  | 14.02% | 0.57 | 0.187600 | 19.91% | 189,055,270 | 47,516,584 | 236,571,854 | 1,261.0439979 |
| 21 | 4,813,820,510 | 9,379,664,542  | 0.15%  | 0.51 | 1.326400 | 0.95%  | 189,055,270 | 47,516,584 | 236,571,854 | 178.3563435   |
| 22 | 4,906,832,629 | 9,633,705,339  | 0.10%  | 0.51 | 1.511800 | 1.55%  | 189,055,270 | 47,516,584 | 236,571,854 | 156.4835653   |
| 23 | 4,992,765,915 | 9,940,291,302  | 0.16%  | 0.50 | 1.541600 | 2.07%  | 189,055,270 | 47,516,584 | 236,571,854 | 153.4586495   |
| 24 | 5,071,148,290 | 10,165,856,353 | 0.18%  | 0.50 | 1.546000 | 1.59%  | 189,055,270 | 47,516,584 | 236,571,854 | 153.0218978   |
| 25 | 5,147,308,380 | 10,432,036,750 | 0.36%  | 0.49 | 1.569600 | 2.47%  | 189,055,270 | 47,516,584 | 236,571,854 | 150.7211098   |

### Results from varying Number of Threads on Intel Core i7-9700 machine

For ease of organisation and to avoid timeout, batch script was submitted in 3 batches (1-10 threads, 11-20 threads, 21-25 threads)

The batch for 1-10 threads was run on pdc-020.

The batch for 11-20 threads was run on pdc-020.

The batch for 21-25 threads was run on pdc-020.

| Number of Threads | Number of Instructions | Number of Cycles | Number of Cycles (Error) | Instructions per cycle (IPC) | Execution Time in seconds | Execution Time (Error) | Number of retired scalar double-precision floating-point instructions | Number of retired 128-bit packed double-precision floating-point instructions | Total number of double-precision floating-point instructions | Number (in millions) of floating-point operations per second (MFLOPS) |
|-------------------|------------------------|------------------|--------------------------|------------------------------|---------------------------|------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------|
| 1                 | 3,437,120,023          | 2,937,510,410    | 0.40%                    | 1.17                         | 0.644060                  | 1.19%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 367.3133776                                                           |
| 2                 | 3,451,736,690          | 3,267,044,333    | 0.66%                    | 1.06                         | 0.363260                  | 0.67%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 651.2466388                                                           |
| 3                 | 3,458,939,369          | 3,392,596,315    | 0.44%                    | 1.02                         | 0.256837                  | 0.37%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 921.0972484                                                           |
| 4                 | 3,464,090,749          | 3,436,384,349    | 0.26%                    | 1.01                         | 0.200820                  | 0.33%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 1,178.0293497                                                         |
| 5                 | 3,469,492,739          | 3,504,567,459    | 0.14%                    | 0.99                         | 0.166930                  | 0.72%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 1,417.1919607                                                         |
| 6                 | 3,476,899,780          | 3,597,840,535    | 0.55%                    | 0.97                         | 0.147090                  | 2.59%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 1,608.3476375                                                         |
| 7                 | 3,480,839,291          | 3,601,510,007    | 0.07%                    | 0.97                         | 0.131450                  | 3.95%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 1,799.7098060                                                         |
| 8                 | 3,500,088,579          | 3,931,222,596    | 2.08%                    | 0.89                         | 0.126730                  | 1.78%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 1,866.7391620                                                         |
| 9                 | 4,002,720,302          | 5,012,969,200    | 0.07%                    | 0.80                         | 0.309100                  | 0.42%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 765.3570171                                                           |
| 10                | 4,066,313,934          | 5,181,900,638    | 0.03%                    | 0.78                         | 0.288620                  | 0.99%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 819.6654910                                                           |
| 11                | 4,149,053,469          | 5,409,101,572    | 0.05%                    | 0.77                         | 0.294780                  | 1.40%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 802.5369903                                                           |
| 12                | 4,248,140,672          | 5,695,079,040    | 0.09%                    | 0.75                         | 0.358510                  | 1.36%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 659.8751890                                                           |
| 13                | 4,364,820,219          | 6,012,292,034    | 0.03%                    | 0.73                         | 0.443930                  | 0.60%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 532.9035073                                                           |
| 14                | 4,503,552,474          | 6,353,798,564    | 0.04%                    | 0.71                         | 0.468350                  | 0.47%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 505.1176556                                                           |

|    |               |               |       |      |          |       |             |            |             |             |
|----|---------------|---------------|-------|------|----------|-------|-------------|------------|-------------|-------------|
| 15 | 4,612,508,720 | 6,626,069,904 | 0.04% | 0.70 | 0.421950 | 0.98% | 189,055,270 | 47,516,584 | 236,571,854 | 560.6632397 |
| 16 | 4,719,303,004 | 6,868,851,738 | 0.06% | 0.69 | 0.374920 | 0.97% | 189,055,270 | 47,516,584 | 236,571,854 | 630.9928891 |
| 17 | 4,814,430,779 | 7,160,939,805 | 0.04% | 0.67 | 0.424530 | 1.67% | 189,055,270 | 47,516,584 | 236,571,854 | 557.2559160 |
| 18 | 4,952,862,067 | 7,585,118,086 | 0.22% | 0.65 | 0.741200 | 4.37% | 189,055,270 | 47,516,584 | 236,571,854 | 319.1741149 |
| 19 | 5,040,747,852 | 7,706,020,044 | 0.05% | 0.65 | 0.409300 | 3.11% | 189,055,270 | 47,516,584 | 236,571,854 | 577.9913364 |
| 20 | 5,171,818,706 | 8,031,563,197 | 0.03% | 0.64 | 0.557630 | 1.13% | 189,055,270 | 47,516,584 | 236,571,854 | 424.2452056 |
| 21 | 5,290,812,370 | 8,264,350,039 | 0.04% | 0.64 | 0.554820 | 1.64% | 189,055,270 | 47,516,584 | 236,571,854 | 426.3938827 |
| 22 | 5,395,377,477 | 8,536,810,045 | 0.04% | 0.63 | 0.487430 | 0.94% | 189,055,270 | 47,516,584 | 236,571,854 | 485.3452886 |
| 23 | 5,512,146,306 | 8,802,609,467 | 0.10% | 0.63 | 0.553660 | 1.33% | 189,055,270 | 47,516,584 | 236,571,854 | 427.2872413 |
| 24 | 5,619,013,025 | 9,075,132,440 | 0.09% | 0.62 | 0.527600 | 2.10% | 189,055,270 | 47,516,584 | 236,571,854 | 448.3924450 |
| 25 | 5,724,863,676 | 9,341,267,496 | 0.10% | 0.61 | 0.579500 | 4.26% | 189,055,270 | 47,516,584 | 236,571,854 | 408.2344331 |

### Results from varying Number of Threads on Intel Core xw-2245 machine

For ease of organisation and to avoid timeout, batch script was submitted in 3 batches (1-10 threads, 11-20 threads, 21-25 threads)

The batch for 1-10 threads was run on pdc-023.

The batch for 11-20 threads was run on pdc-023.

The batch for 21-25 threads was run on pdc-023.

| Number of Threads | Number of Instructions | Number of Cycles | Number of Cycles (Error) | Instructions per cycle (IPC) | Execution Time in seconds | Execution Time (Error) | Number of retired scalar double-precision floating-point instructions | Number of retired 128-bit packed double-precision floating-point instructions | Total number of double-precision floating-point instructions | Number (in millions) of floating-point operations per second (MFLOPS) |
|-------------------|------------------------|------------------|--------------------------|------------------------------|---------------------------|------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------|
| 1                 | 3,437,325,924          | 2,822,401,378    | 1.25%                    | 1.22                         | 0.638200                  | 2.49%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 370.6860765                                                           |
| 2                 | 3,504,379,456          | 3,354,333,903    | 0.42%                    | 1.04                         | 0.409120                  | 1.04%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 578.2456345                                                           |
| 3                 | 3,544,124,828          | 3,663,828,096    | 2.15%                    | 0.97                         | 0.309330                  | 2.93%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 764.7879417                                                           |
| 4                 | 3,534,271,253          | 3,587,117,085    | 3.60%                    | 0.99                         | 0.245200                  | 8.06%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 964.8118026                                                           |
| 5                 | 3,574,543,484          | 3,842,127,537    | 2.20%                    | 0.93                         | 0.203610                  | 1.80%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 1,161.8872059                                                         |
| 6                 | 3,593,497,702          | 3,981,521,861    | 2.10%                    | 0.90                         | 0.181100                  | 7.71%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 1,306.3051022                                                         |
| 7                 | 3,602,879,961          | 4,006,098,944    | 3.52%                    | 0.90                         | 0.164500                  | 9.92%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 1,438.1267720                                                         |
| 8                 | 3,628,833,845          | 4,213,899,171    | 0.67%                    | 0.86                         | 0.141300                  | 8.77%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 1,674.2523284                                                         |
| 9                 | 3,688,378,475          | 4,786,081,331    | 1.65%                    | 0.77                         | 0.149400                  | 11.62%                 | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 1,583.4796118                                                         |
| 10                | 3,725,846,759          | 5,365,637,594    | 0.91%                    | 0.69                         | 0.138570                  | 4.91%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 1,707.2371653                                                         |
| 11                | 3,719,763,999          | 5,565,608,633    | 2.25%                    | 0.67                         | 0.144100                  | 10.86%                 | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 1,641.7200139                                                         |
| 12                | 3,758,621,240          | 5,999,459,282    | 0.14%                    | 0.63                         | 0.132650                  | 3.66%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 1,783.4289785                                                         |
| 13                | 3,767,105,645          | 6,182,495,265    | 3.02%                    | 0.61                         | 0.139300                  | 13.37%                 | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 1,698.2904092                                                         |
| 14                | 3,765,106,123          | 6,256,195,390    | 1.67%                    | 0.60                         | 0.128400                  | 8.92%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 1,842.4599221                                                         |

|    |               |               |       |      |          |       |             |            |             |               |
|----|---------------|---------------|-------|------|----------|-------|-------------|------------|-------------|---------------|
| 15 | 3,910,341,642 | 7,538,111,178 | 3.37% | 0.52 | 0.139530 | 4.86% | 189,055,270 | 47,516,584 | 236,571,854 | 1,695.4909625 |
| 16 | 4,094,848,740 | 8,945,324,880 | 2.03% | 0.46 | 0.150300 | 7.17% | 189,055,270 | 47,516,584 | 236,571,854 | 1,573.9976979 |
| 17 | 4,563,331,892 | 6,607,755,461 | 0.51% | 0.69 | 0.977600 | 5.30% | 189,055,270 | 47,516,584 | 236,571,854 | 241.9924857   |
| 18 | 4,743,086,562 | 7,068,502,356 | 0.61% | 0.67 | 1.169500 | 4.27% | 189,055,270 | 47,516,584 | 236,571,854 | 202.2846122   |
| 19 | 4,780,091,724 | 7,167,311,790 | 0.68% | 0.67 | 0.962900 | 3.18% | 189,055,270 | 47,516,584 | 236,571,854 | 245.6868356   |
| 20 | 4,901,775,262 | 7,415,325,494 | 0.06% | 0.66 | 1.078800 | 5.49% | 189,055,270 | 47,516,584 | 236,571,854 | 219.2916704   |
| 21 | 4,989,571,243 | 7,596,059,783 | 1.11% | 0.66 | 1.010700 | 2.40% | 189,055,270 | 47,516,584 | 236,571,854 | 234.0673335   |
| 22 | 5,054,916,286 | 7,979,778,668 | 0.63% | 0.63 | 1.049100 | 8.33% | 189,055,270 | 47,516,584 | 236,571,854 | 225.4998132   |
| 23 | 5,115,863,911 | 8,137,904,930 | 0.21% | 0.63 | 1.007500 | 5.69% | 189,055,270 | 47,516,584 | 236,571,854 | 234.8107732   |
| 24 | 5,235,675,168 | 8,247,265,182 | 0.30% | 0.63 | 1.072200 | 2.26% | 189,055,270 | 47,516,584 | 236,571,854 | 220.6415352   |
| 25 | 5,325,860,384 | 8,330,575,425 | 1.14% | 0.64 | 1.127300 | 6.83% | 189,055,270 | 47,516,584 | 236,571,854 | 209.8570514   |



### Results from varying Number of Particles on Intel Core i7-7700 machine

| Number of Particles | Number of Instructions | Number of Cycles | Number of Cycles (Error) | Instructions per cycle (IPC) | Execution Time in seconds | Execution Time (Error) | Number of retired scalar double-precision floating-point instructions | Number of retired 128-bit packed double-precision floating-point instructions | Total number of double-precision floating-point instructions | Number (in millions) of floating-point operations per second (MFLOPS) |
|---------------------|------------------------|------------------|--------------------------|------------------------------|---------------------------|------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------|
| 4000                | 566,155,505            | 811,765,606      | 0.70%                    | 0.70                         | 0.039350                  | 21.47%                 | 9,875,195                                                             | 2,810,999                                                                     | 12,686,194                                                   | 322.3937484                                                           |
| 5000                | 736,423,059            | 1,231,630,074    | 3.68%                    | 0.60                         | 0.049790                  | 10.94%                 | 16,919,802                                                            | 4,684,002                                                                     | 21,603,804                                                   | 433.8984535                                                           |
| 6000                | 954,999,585            | 1,379,955,089    | 11.09%                   | 0.69                         | 0.067100                  | 28.92%                 | 27,747,364                                                            | 7,543,193                                                                     | 35,290,557                                                   | 525.9397466                                                           |
| 7000                | 1,214,124,411          | 1,733,454,019    | 0.28%                    | 0.70                         | 0.071470                  | 6.67%                  | 42,352,479                                                            | 11,315,889                                                                    | 53,668,368                                                   | 750.9216175                                                           |
| 8000                | 1,607,490,722          | 2,266,827,091    | 0.17%                    | 0.71                         | 0.084540                  | 4.28%                  | 65,493,546                                                            | 17,260,124                                                                    | 82,753,670                                                   | 978.8700024                                                           |
| 9000                | 2,359,827,917          | 3,302,705,330    | 0.84%                    | 0.71                         | 0.121920                  | 5.53%                  | 111,356,321                                                           | 28,704,596                                                                    | 140,060,917                                                  | 1,148.7936106                                                         |
| 10000               | 3,503,987,909          | 5,339,584,677    | 1.70%                    | 0.66                         | 0.188800                  | 6.89%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 1,253.0288877                                                         |

### Results from varying Grid Size on Intel Core i7-7700 machine

| Grid Size | Number of Instructions | Number of Cycles | Number of Cycles (Error) | Instructions per cycle (IPC) | Execution Time in seconds | Execution Time (Error) | Number of retired scalar double-precision floating-point instructions | Number of retired 128-bit packed double-precision floating-point instructions | Total number of double-precision floating-point instructions | Number (in millions) of floating-point operations per second (MFLOPS) |
|-----------|------------------------|------------------|--------------------------|------------------------------|---------------------------|------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------|
| 10000     | 3,490,933,360          | 5,083,150,279    | 0.29%                    | 0.69                         | 0.175290                  | 2.37%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 1,349.6026813                                                         |
| 20000     | 1,530,824,301          | 2,009,196,548    | 0.18%                    | 0.76                         | 0.078310                  | 3.57%                  | 12,985,391                                                            | 4,183,028                                                                     | 17,168,419                                                   | 219.2366109                                                           |
| 30000     | 2,179,597,200          | 2,432,990,488    | 0.16%                    | 0.90                         | 0.096290                  | 5.59%                  | 5,846,320                                                             | 2,140,302                                                                     | 7,986,622                                                    | 82.9434209                                                            |
| 40000     | 3,048,668,437          | 3,199,133,221    | 2.87%                    | 0.95                         | 0.135200                  | 16.28%                 | 3,831,531                                                             | 1,546,240                                                                     | 5,377,771                                                    | 39.7764127                                                            |
| 50000     | 4,246,067,762          | 4,472,884,986    | 0.59%                    | 0.95                         | 0.159210                  | 2.36%                  | 3,104,490                                                             | 1,328,272                                                                     | 4,432,762                                                    | 27.8422335                                                            |
| 60000     | 5,879,956,715          | 6,806,260,776    | 2.05%                    | 0.86                         | 0.229230                  | 2.29%                  | 2,772,037                                                             | 1,227,871                                                                     | 3,999,908                                                    | 17.4493216                                                            |
| 70000     | 7,442,891,959          | 9,022,971,316    | 0.11%                    | 0.82                         | 0.310150                  | 1.19%                  | 2,550,269                                                             | 1,159,501                                                                     | 3,709,770                                                    | 11.9612123                                                            |

### Results from varying Particle Radius on Intel Core i7-7700 machine

| Particle Radius | Number of Instructions | Number of Cycles | Number of Cycles (Error) | Instructions per cycle (IPC) | Execution Time in seconds | Execution Time (Error) | Number of retired scalar floating-point instructions | Number of retired 128-bit packed double-precision floating-point instructions | Total number of double-precision floating-point instructions | Number (in millions) of floating-point operations per second (MFLOPS) |
|-----------------|------------------------|------------------|--------------------------|------------------------------|---------------------------|------------------------|------------------------------------------------------|-------------------------------------------------------------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------|
| 47              | 2,635,380,851          | 3,768,574,959    | 4.14%                    | 0.70                         | 0.139700                  | 7.32%                  | 122,274,425                                          | 31,712,199                                                                    | 153,986,624                                                  | 1,102.2664567                                                         |
| 48              | 2,860,263,647          | 4,087,758,404    | 0.44%                    | 0.70                         | 0.142610                  | 1.13%                  | 139,151,921                                          | 35,793,363                                                                    | 174,945,284                                                  | 1,226.7392469                                                         |
| 49              | 3,063,352,110          | 4,387,115,701    | 3.79%                    | 0.70                         | 0.159500                  | 6.46%                  | 157,205,336                                          | 39,963,629                                                                    | 197,168,965                                                  | 1,236.1690596                                                         |
| 50              | 3,512,376,113          | 5,552,929,215    | 2.76%                    | 0.63                         | 0.179640                  | 3.68%                  | 189,055,270                                          | 47,516,584                                                                    | 236,571,854                                                  | 1,316.9219216                                                         |
| 51              | 4,462,159,566          | 6,621,532,884    | 0.39%                    | 0.67                         | 0.230500                  | 5.45%                  | 257,018,511                                          | 63,424,432                                                                    | 320,442,943                                                  | 1,390.2079957                                                         |
| 52              | 9,233,898,180          | 14,291,234,143   | 0.40%                    | 0.65                         | 0.472870                  | 1.45%                  | 577,163,538                                          | 139,559,165                                                                   | 716,722,703                                                  | 1,515.6865587                                                         |
| 53              | 16,332,413,961         | 25,095,806,259   | 1.24%                    | 0.65                         | 0.824200                  | 1.42%                  | 1,092,713,527                                        | 267,925,399                                                                   | 1,360,638,926                                                | 1,650.8601383                                                         |

### Results from varying Number of Timesteps on Intel Core i7-7700 machine

| Number of Timesteps | Number of Instructions | Number of Cycles | Number of Cycles (Error) | Instructions per cycle (IPC) | Execution Time in seconds | Execution Time (Error) | Number of retired scalar double-precision floating-point instructions | Number of retired 128-bit packed double-precision floating-point instructions | Total number of double-precision floating-point instructions | Number (in millions) of floating-point operations per second (MFLOPS) |
|---------------------|------------------------|------------------|--------------------------|------------------------------|---------------------------|------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------|
| 100                 | 3,491,350,744          | 5,072,408,370    | 2.05%                    | 0.69                         | 0.209300                  | 15.68%                 | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 1,130.3003058                                                         |
| 200                 | 6,325,955,122          | 9,339,095,238    | 1.24%                    | 0.68                         | 0.350300                  | 10.63%                 | 339,185,757                                                           | 87,768,466                                                                    | 426,954,223                                                  | 1,218.8245019                                                         |
| 300                 | 8,175,699,762          | 11,720,885,065   | 0.89%                    | 0.70                         | 0.434800                  | 10.83%                 | 430,109,093                                                           | 113,904,654                                                                   | 544,013,747                                                  | 1,251.1815708                                                         |
| 400                 | 9,910,431,343          | 14,664,839,810   | 0.95%                    | 0.68                         | 0.520500                  | 7.51%                  | 511,097,172                                                           | 137,612,365                                                                   | 648,709,537                                                  | 1,246.3199558                                                         |
| 500                 | 11,581,526,787         | 16,602,226,302   | 0.54%                    | 0.70                         | 0.584900                  | 8.84%                  | 590,810,966                                                           | 161,004,169                                                                   | 751,815,135                                                  | 1,285.3737989                                                         |
| 600                 | 13,388,365,599         | 19,730,933,799   | 1.63%                    | 0.68                         | 0.663600                  | 6.84%                  | 676,169,319                                                           | 185,708,447                                                                   | 861,877,766                                                  | 1,298.7910880                                                         |
| 700                 | 15,269,807,238         | 21,365,289,106   | 1.34%                    | 0.71                         | 0.750500                  | 7.53%                  | 769,775,104                                                           | 212,469,728                                                                   | 982,244,832                                                  | 1,308.7872512                                                         |
| 800                 | 16,942,390,440         | 24,055,874,776   | 1.31%                    | 0.70                         | 0.826500                  | 6.56%                  | 848,161,764                                                           | 235,427,272                                                                   | 1,083,589,036                                                | 1,311.0575148                                                         |
| 900                 | 18,499,183,340         | 25,963,471,054   | 0.71%                    | 0.71                         | 0.884400                  | 4.84%                  | 921,911,739                                                           | 257,234,970                                                                   | 1,179,146,709                                                | 1,333.2730767                                                         |
| 1000                | 20,278,419,202         | 29,091,995,922   | 0.85%                    | 0.70                         | 0.960800                  | 4.36%                  | 1,005,061,453                                                         | 281,397,600                                                                   | 1,286,459,053                                                | 1,338.9457254                                                         |

Results from varying **CELL\_SIZE\_MULTIPLIER** on Intel Core i7-7700 machine

| CELL_SIZE_MULTIPLIER | Number of Instructions | Number of Cycles | Number of Cycles (Error) | Instructions per cycle (IPC) | Execution Time in seconds | Execution Time (Error) | Number of retired scalar double-precision floating-point instructions | Number of retired 128-bit packed double-precision floating-point instructions | Total number of double-precision floating-point instructions | Number (in millions) of floating-point operations per second (MFLOPS) |
|----------------------|------------------------|------------------|--------------------------|------------------------------|---------------------------|------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------|
| 2.0                  | 3,490,495,544          | 5,060,513,858    | 2.80%                    | 0.69                         | 0.179060                  | 5.23%                  | 189,055,270                                                           | 47,516,585                                                                    | 236,571,855                                                  | 1,321.1876187                                                         |
| 2.5                  | 5,114,758,872          | 6,840,181,416    | 2.62%                    | 0.75                         | 0.255700                  | 8.15%                  | 467,349,766                                                           | 134,925,023                                                                   | 602,274,789                                                  | 2,355.3961244                                                         |
| 3.0                  | 7,842,760,196          | 8,877,062,321    | 0.20%                    | 0.88                         | 0.306100                  | 5.29%                  | 1,006,673,312                                                         | 313,582,562                                                                   | 1,320,255,874                                                | 4,313.1521529                                                         |
| 3.5                  | 7,328,060,375          | 8,007,584,339    | 1.72%                    | 0.92                         | 0.271500                  | 1.49%                  | 994,198,023                                                           | 314,232,149                                                                   | 1,308,430,172                                                | 4,819.2639853                                                         |
| 4.0                  | 3,499,297,291          | 5,252,142,037    | 5.17%                    | 0.67                         | 0.184630                  | 5.12%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 1,281.3294373                                                         |
| 4.5                  | 3,513,432,408          | 5,554,273,127    | 2.59%                    | 0.63                         | 0.177230                  | 2.91%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 1,334.8296225                                                         |
| 5.0                  | 3,491,962,047          | 5,103,271,057    | 0.72%                    | 0.68                         | 0.172340                  | 0.75%                  | 189,055,270                                                           | 47,516,584                                                                    | 236,571,854                                                  | 1,372.7042706                                                         |
| 5.5                  | 10,981,351,112         | 10,738,571,103   | 1.70%                    | 1.02                         | 0.369500                  | 4.24%                  | 1,637,318,760                                                         | 524,694,610                                                                   | 2,162,013,370                                                | 5,851.1863870                                                         |
| 6.0                  | 13,628,481,960         | 12,300,894,287   | 1.00%                    | 1.11                         | 0.457800                  | 3.62%                  | 2,154,418,948                                                         | 697,089,170                                                                   | 2,851,508,118                                                | 6,228.7202228                                                         |
| 6.5                  | 13,664,048,411         | 13,089,983,786   | 2.02%                    | 1.04                         | 0.464500                  | 1.93%                  | 2,154,418,948                                                         | 697,089,170                                                                   | 2,851,508,118                                                | 6,138.8764650                                                         |
| 7.0                  | 13,643,298,131         | 12,654,584,418   | 0.40%                    | 1.08                         | 0.534900                  | 3.15%                  | 2,154,418,948                                                         | 697,089,170                                                                   | 2,851,508,118                                                | 5,330.9181492                                                         |

### Results from varying on Intel Core i7-7700 machine: Comparison between Grid Adjacency Conversion

| Thread<br>s | Subgrid<br>Size | Number of<br>Instructions | Number of<br>Cycles | Number<br>of<br>Cycles<br>(Error) | Instructions<br>per cycle<br>(IPC) | Execution<br>Time in<br>seconds | Execution<br>Time<br>(Error) | Cache<br>References | Cache<br>References<br>(Error) | Cache<br>Misses | Cache<br>Misses<br>(Error) | Cache<br>Miss<br>Rate |
|-------------|-----------------|---------------------------|---------------------|-----------------------------------|------------------------------------|---------------------------------|------------------------------|---------------------|--------------------------------|-----------------|----------------------------|-----------------------|
| 1           | 2x2             | 4,693,586,784             | 3,616,362,611       | 0.00%                             | 1.29                               | 1.282500                        | 0.01%                        | 25,197,808          | 5.49%                          | 41,392          | 1.34%                      | 0.15%                 |
| 2           | 2x2             | 4,713,028,372             | 3,994,562,801       | 0.02%                             | 1.18                               | 0.757100                        | 0.01%                        | 13,853,423          | 3.54%                          | 43,023          | 1.11%                      | 0.29%                 |
| 3           | 2x2             | 4,720,566,174             | 4,163,846,216       | 0.02%                             | 1.12                               | 0.556400                        | 0.01%                        | 17,720,469          | 0.82%                          | 36,150          | 1.51%                      | 0.21%                 |
| 4           | 2x2             | 4,739,967,086             | 4,635,387,153       | 0.07%                             | 1.07                               | 0.460000                        | 0.01%                        | 19,595,489          | 1.30%                          | 45,955          | 4.56%                      | 0.23%                 |
| 5           | 2x2             | 4,739,656,426             | 4,477,245,227       | 0.01%                             | 1.06                               | 0.383290                        | 0.00%                        | 21,124,150          | 0.25%                          | 28,213          | 7.31%                      | 0.13%                 |
| 6           | 2x2             | 4,745,288,978             | 4,530,485,154       | 0.01%                             | 1.05                               | 0.333100                        | 0.00%                        | 21,601,146          | 0.07%                          | 30,305          | 5.43%                      | 0.14%                 |
| 7           | 2x2             | 4,757,172,460             | 4,720,079,089       | 0.03%                             | 1.01                               | 0.296200                        | 0.01%                        | 21,863,085          | 0.20%                          | 23,957          | 21.12%                     | 0.11%                 |
| 8           | 2x2             | 4,761,594,839             | 4,736,943,287       | 0.01%                             | 1.00                               | 0.282278                        | 0.00%                        | 22,158,777          | 0.25%                          | 24,246          | 20.37%                     | 0.11%                 |
| 9           | 2x2             | 4,768,224,323             | 4,813,083,723       | 0.01%                             | 0.99                               | 0.249000                        | 0.01%                        | 22,660,120          | 0.48%                          | 21,295          | 20.83%                     | 0.09%                 |
| 10          | 2x2             | 4,776,322,922             | 4,929,572,417       | 0.03%                             | 0.97                               | 0.232000                        | 0.01%                        | 23,144,857          | 0.39%                          | 22,463          | 18.32%                     | 0.10%                 |
| 1           | 2x3             | 4,597,988,312             | 3,942,943,626       | 0.00%                             | 1.16                               | 1.460800                        | 0.05%                        | 33,847,411          | 2.47%                          | 38,972          | 2.46%                      | 0.1100<br>00%         |
| 2           | 2x3             | 4,621,709,130             | 4,334,038,218       | 0.02%                             | 1.07                               | 0.808840                        | 0.01%                        | 15,617,122          | 1.56%                          | 42,111          | 1.48%                      | 0.2690<br>00%         |
| 3           | 2x3             | 4,633,838,182             | 4,574,823,531       | 0.00%                             | 1.01                               | 0.603800                        | 0.01%                        | 19,345,861          | 0.92%                          | 40,873          | 8.82%                      | 0.2100<br>00%         |
| 4           | 2x3             | 4,644,127,133             | 4,759,856,240       | 0.01%                             | 0.97                               | 0.493430                        | 0.00%                        | 21,694,475          | 0.63%                          | 31,500          | 7.49%                      | 0.1450<br>00%         |
| 5           | 2x3             | 4,656,047,193             | 4,928,690,745       | 0.01%                             | 0.95                               | 0.419300                        | 0.01%                        | 23,043,389          | 0.47%                          | 31,941          | 11.68%                     | 0.1400                |

|    |     |               |               |       |      |          |       |            |       |            |       |                |
|----|-----|---------------|---------------|-------|------|----------|-------|------------|-------|------------|-------|----------------|
|    |     |               |               |       |      |          |       |            |       |            |       | 00%            |
| 6  | 2x3 | 4,687,359,289 | 6,861,356,291 | 0.05% | 0.69 | 0.465360 | 0.01% | 24,733,534 | 0.32% | 13,660,656 | 1.55% | 54.991<br>000% |
| 7  | 2x3 | 4,698,636,334 | 7,010,529,481 | 0.01% | 0.67 | 0.406600 | 0.01% | 25,874,742 | 0.19% | 14,150,509 | 0.25% | 54.499<br>000% |
| 8  | 2x3 | 4,709,808,862 | 7,191,138,985 | 0.14% | 0.64 | 0.390470 | 0.01% | 26,755,578 | 0.43% | 14,069,170 | 0.57% | 52.672<br>000% |
| 9  | 2x3 | 4,729,384,498 | 7,603,553,371 | 0.08% | 0.62 | 0.368490 | 0.01% | 28,284,888 | 0.65% | 14,202,977 | 1.33% | 50.838<br>000% |
| 10 | 2x3 | 4,750,924,447 | 8,090,705,203 | 0.01% | 0.59 | 0.349000 | 0.01% | 29,360,955 | 1.25% | 15,015,542 | 1.53% | 51.697<br>000% |